STAT 210A: Theoretical Statistics

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Fall 2026

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11/5/26 Lecture 21: Monotone Likelihood Ratios, UMP Tests, and the Score Test

Example: Binomial

Consider $X \sim \text{Binom}(n, \theta)$, $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$. Recall the LRT test:

$$\phi_\alpha^*(x) = \begin{cases} 1 & \text{LR}(x) > c_\alpha \\ \gamma_\alpha & \text{LR}(x) = c_\alpha \\ 0 & \text{LR}(x) < c_\alpha \end{cases}$$

where we choose c_α and γ_α so $\mathbb{E}_{H_0}\phi_\alpha(X) = \alpha$.

Because the test didn't end up depending on the value of θ_1, just that $\theta_1 > \theta_0$, we see the LRT works for

$$H_0 : \theta = \theta_0 \qquad H_1 : \theta > \theta_0$$

and as it turns out, even for

$$H_0 : \theta \leq \theta_0 \qquad H_1 : \theta > \theta_0$$

Recall

$$\text{LR}(x) = \frac{\theta_1^x(1-\theta_1)^{n-x}}{\theta_0^x(1-\theta_0)^{n-x}} \propto_x \left(\frac{\theta_1}{1-\theta_1} \cdot \frac{1-\theta_0}{\theta_0}\right)^x$$

which is increasing in x if $\theta_1 > \theta_0$. We say LR is monotone in x.

Definition 21.1: Monotone Likelihood Ratio

Assume $\mathcal{P} = \{P_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$ has densities p_θ. We say $\mathcal{P}$ has *monotone likelihood ratio (MLR)* in a test statistic $T(x)$ if

$$\frac{p_{\theta_2}}{p_{\theta_1}}(x) \text{ is a nondecreasing function of } T(x)$$

whenever $\theta_1 < \theta_2$.

Example: 1-Parameter Exponential Families

Let $X \sim p_\eta(x) = e^{\eta T(x) - A(\eta)}h(x)$. Then,

$$\frac{p_{\eta_2}}{p_{\eta_1}}(x) = \exp\{(\eta_2 - \eta_1)T(x) - (A(\eta_2) - A(\eta_1))\}$$

so if $\eta_2 > \eta_1$, we see LR is increasing in $T(x)$, so the LRT for $H_0 : \eta = \eta_0$ vs. $H_1 : \eta = \eta_1$ with $\eta_1 > \eta_0$ will reject for large $T(x)$.

Definition 21.2: Uniformly Most Powerful Tests

If ϕ^* is level α and for any other level-α test ϕ,

$$\mathbb{E}_\theta \phi^*(X) \geq \mathbb{E}_\theta \phi(X) \quad \text{for all } \theta \in \Theta_1$$

then ϕ^* is *uniformly most powerful* or *UMP* for that testing problem.

Theorem 21.3

Assume $\mathcal{P}$ has MLR in $T(X)$ and consider testing

$$H_0 : \theta \leq \theta_0 \quad \text{and} \quad H_1 : \theta > \theta_0 \quad \text{for } \theta_0 \in \Theta^\circ \subseteq \mathbb{R}$$

If $\phi^*(X)$ rejects for large $T(X)$, then ϕ^* is UMP at level $\alpha = \mathbb{E}_{\theta_0}\phi^*(X)$.

Proof: Let ϕ be another level-α test and let $\theta_1 > \theta_0$. Then, ϕ is a level-α test for testing $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$.

Since $\phi^*(X)$ rejects for large $T(X)$, it is an LR test, so by the Neyman-Pearson Theorem, $\mathbb{E}_{\theta_1}\phi^* \geq \mathbb{E}_{\theta_1}\phi$. Thus, ϕ^* is UMP for $H_0 : \theta = \theta_0$ and $H_1 : \theta > \theta_0$. We now need to make sure it controls the power on H_0.

Note that we have $\mathbb{E}_{\theta_1}\phi^*(X) \geq \alpha$ on H_1 by testing against $\phi(X) \equiv \alpha$.

Now, consider testing $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_2$, $\theta_2 < \theta_0$. Define

$$\overline{\phi}(x) = 1 - \phi^*(x)$$

This test rejects for small T, and is level $(1-\alpha)$. This is an LRT for this hypothesis, which implies that, by Neyman-Pearson, testing it against $\phi(x) \equiv 1 - \alpha$,

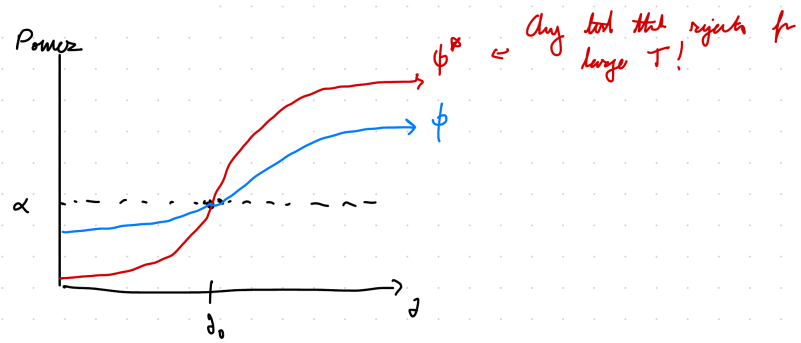
$$\mathbb{E}_{\theta_2}[\overline{\phi}(X)] \geq 1 - \alpha \implies \mathbb{E}_{\theta_2}\phi^*(X) \leq \alpha$$

which was to be shown as $\theta_2 \in H_0$. Even stronger, note that for any other $\phi(x)$ with $\mathbb{E}_{\theta_0}\phi(x) = \alpha$, then testing $1 - \phi(x)$ against $\overline{\phi}(x)$, we get

$$\mathbb{E}_{\theta_2}\phi^*(X) \leq \mathbb{E}_{\theta_2}\phi(X)$$

so ϕ^* is even better than ϕ on H_0!

■



One Parameter Families

Let $\mathcal{P} = \{P_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$. Consider

$$H_0 : \theta \leq \theta_0 \quad \text{vs} \quad H_1 : \theta > \theta_0$$

We call this a *one-sided hypothesis test*. We say a statistic $T(X)$ is *stochastically increasing* in θ if

$$\mathbb{P}_\theta(T(X) \leq t) \text{ is non-increasing in } \theta, \text{ for all } t$$

(or $\mathbb{P}_\theta(T(X) > t)$ is non-decreasing in θ).

If $\phi(x)$ rejects for large $T(x)$ and $\mathbb{E}_{\theta_0}\phi(X) = \alpha$, then

$$\begin{aligned}\mathbb{E}_\theta\phi(X) &= \mathbb{P}_\theta(T > c) + \gamma\mathbb{P}_\theta(T = c) \\ &= (1-\gamma)\mathbb{P}_\theta(T > c) + \gamma\mathbb{P}_\theta(T \geq c)\end{aligned}$$

so as these are both non-decreasing in θ, $\mathbb{E}_\theta\phi(X)$ is increasing in θ, so it is level-α and we can calibrate at the boundary.

Heuristic: We want to maximize power for values near the boundary (if a UMP doesn't exist), which motivates the score test.

Score Test

We see

$$\log \mathrm{LR}(X) = \log \frac{p_{\theta_0+\epsilon}(X)}{p_{\theta_0}(X)} \approx \epsilon \cdot \dot{\ell}(\theta_0; X)$$

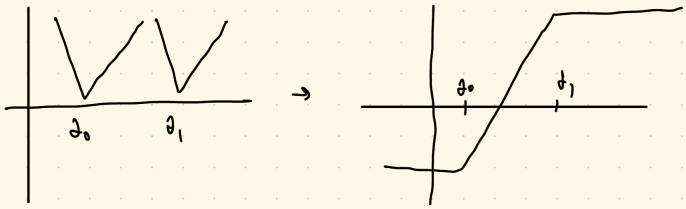
(locally most powerful near θ_0), so we use $S_{\theta_0}(X) = \dot{\ell}(\theta_0; X)$. Then, we reject for large values of the score. Here is a nice example.

Example: Laplace

Consider Laplace $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \frac{1}{2}e^{-|x-\theta|}$. We want to test $H_0 : \theta \leq \theta_0$ vs. $H_1 : \theta > \theta_0$. For $\theta_1 > \theta_0$,

$$\begin{aligned}\log \frac{p_{\theta_1}(x)}{p_{\theta_0}(x)} &= \sum_{i=1}^n (|x_i - \theta_0| - |x_i - \theta_1|) \\ &= \sum_{i=1}^n T(x_i; \theta_0, \theta_1)\end{aligned}$$

What does T look like?



so

$$T(x) = \begin{cases} \theta_0 - \theta_1 & x < \theta_0 \\ 2x - \theta_0 - \theta_1 & \theta_0 \le x \le \theta_1 \\ \theta_1 - \theta_0 & x \ge \theta_1 \end{cases}$$

Note, taking $\theta_1 \to \theta_0$, we get a jump function, and renormalizing,

$$\tilde{T}(x) = \begin{cases} +1 & x > \theta_1 \\ \frac{2x-\theta_0-\theta_1}{\theta_1-\theta_0} & \theta_0 \le x \le \theta_1 \\ -1 & x < \theta_0 \end{cases} \to S_{\theta_0}(x) = \operatorname{sgn}(x - \theta_0)$$

so as $\theta_1 \downarrow \theta_0$,

$$\begin{aligned} \frac{1}{\theta_1 - \theta_0} \log \mathrm{LR}(X) \to S_{\theta_0}(X) &= \sum_{i=1}^n \operatorname{sgn}(X_i - \theta_0) \\ &= 2\#\{X_i > \theta_0\} - n \end{aligned}$$

where

$$\begin{aligned} \#\{X_i > \theta_0\} &\sim \mathrm{Binom}(n, \mathbb{P}(X_i > \theta_0)) \\ &= \mathrm{Binom}(n, \mathbb{P}_\theta(Z_i + \theta > \theta_0)) \end{aligned}$$

which is stochastically increasing in θ. We call it the "*sign test*": we reject for the upper α quantile of $\mathrm{Binom}(n, 1/2)$. Let

$$\begin{aligned} B_{\theta_0}(X) = \sum_{i=1}^n 1\{X_i > \theta_0\} &\sim \mathrm{Binom}(n, \mathbb{P}(X_i > \theta_0)) \\ &\overset{\theta=\theta_0}{=} \mathrm{Binom}(n, 1/2) \end{aligned}$$

Example: Quantiles

$X_1, \dots, X_n \overset{\text{iid}}{\sim} F$ (a cdf on $\mathbb{R}$). Define the "lower ζ quantile"

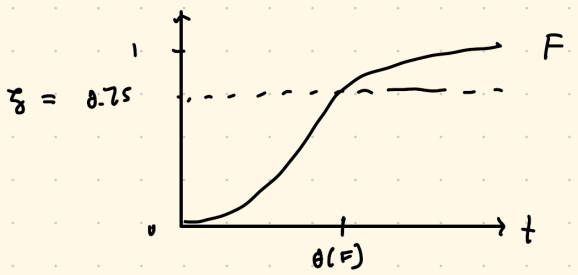
$$\theta(F) = \min\{t : F(t) \ge \zeta\}$$

A minimum is attained since F is right-continuous. We want to test

$$H_0 : \theta(F) \le \theta_0 \quad \text{vs} \quad H_1 : \theta(F) > \theta_0$$

which is equivalent to testing

$$H_0 : F(\theta_0) \ge \zeta \quad \text{vs} \quad H_1 : F(\theta_0) < \zeta$$



It turns out a good test is

$$B_{\theta_0}(X) = \#\{X_i > \theta_0\} \sim \text{Binom}(n, 1 - F(\theta_0)) \overset{H_0}{\leq} \text{Binom}(n, 1 - \zeta)$$

and we reject for large values of B_{θ_0}.